

Dynamic Target Localization and Optical Link Maintenance Using a 2D Beam Scanning System for IoT Connectivity

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I. INTRODUCTION

Abstract—Optical wireless communication (OWC) is gaining momentum as a key enabler for next-generation indoor Internet of Things (IoT) systems, offering high-bandwidth, low-latency, and interference-free connectivity by leveraging its inherent line-of-sight nature. Its versatility makes it attractive for a wide range of dynamic applications, including flexible interconnects in data centers, high-speed data delivery to mobile devices, and communication with cooperative robots. However, conventional solutions still face critical challenges. Visible light communication and positioning (VLCP) is constrained by limited modulation bandwidth and a trade-off between localization precision and data rate, while laser-based OWC systems often suffer from alignment sensitivity, making them less effective in dynamic or mobile scenarios. To overcome these barriers, we present a dynamic localization and two-dimensional (2D) beam scanning communication system based on a single tunable MEMS vertical-cavity surface-emitting laser (MEMS-VCSEL) integrated with a corner cube reflector (CCR). This compact, coaxial optical architecture enables precise real-time tracking and continuous beam alignment without relying on external sensors. A Search-and-Communicate mechanism facilitates autonomous target acquisition and link maintenance. Experimental validation demonstrates sub-3 mm tracking error, stable communication over a 3-meter communication link achieving data rates up to 3.4 Gbps-while the target is in motion.

Index Terms—Automatic alignment, corner cube reflector (CCR), dynamic localization, optical wireless communication (OWC), tunable microelectromechanical system-vertical-cavity surface-emitting laser (MEMS-VCSEL).

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OPTICAL wireless communication (OWC) offers a promising solution to alleviate the congestion of the radio spectrum without necessarily replacing it. Instead, OWC can serve as a valuable complement in various scenarios, such as in environments with high electromagnetic interference (e.g., hospitals and industrial settings). In densely populated urban areas, where the radio frequency (RF) spectrum is heavily utilized, especially in the 2.4 GHz and 5 GHz bands used by Wi-Fi, OWC can offer additional bandwidth. Furthermore, the inherent line-of-sight nature of OWC enhances security and data privacy, making it particularly well-suited for sensitive applications such as military communications and secure financial transactions. The convergence of sensing and communication capabilities within a single framework, termed Integrated Sensing and Communications (ISAC), is poised to become a cornerstone for the development of future wireless networks, including those beyond 5G (B5G) and 6G [1]. This amalgamation is designed to bolster the performance of wireless system. In particular, the optical-ISAC (O-ISAC) is gaining traction in emerging indoor Internet of Things (IoT) scenarios, including data-intensive applications like inter-rack communication in data centers, real-time data transfer for cooperative robots, and low-latency streaming for mobile devices.

For the indoor OWC system, there are two primary approaches: visible light communication (VLC) and beam-steered infrared light communication (BS-ILC), which relies on laser-based transmission [2]. VLC which utilizes LED light sources, is commonly employed in indoor settings but faces limitations due to its narrow bandwidth, typically around 20 MHz, which results in relatively low data rates. Moreover, VLC is often integrated with visible light positioning (VLP), which use the same LED sources for both communication and positioning [3], [4], [5]. Several studies highlight the trade-offs and limitations of VLC systems incorporating VLP. For instance, Yang et al. introduces a differential phase difference of arrival localization method using multiple ceiling-mounted LEDs to support a single mobile receiver over a $2 \times 2 \text{ m}^2$ area [3]. However, the system faces a trade-off between communication and positioning performance—higher communication power improves data rates but reduces positioning precision, and vice

TABLE I
INDOOR OPTICAL WIRELESS COMMUNICATION SYSTEM ADVANCEMENTS

Ref. no.	System type	Distance; Area coverage	No. of light path considered	Main technique	Modulation format	Data rate	Tracked error; Latency	Mobility; Speed
[3]	VLC	1.2-1.8 m; $2 \times 2 \text{ m}^2$	4	Advanced differential phase difference of arrival	2-QAM to 64-QAM	22.3-112 Mbps	8 cm - 0.138 m; N/R	Yes; N/R
[4]		2 m; N/R	256	Combinatorial light emission	4-PAM and 8-PAM	80 kbps	N/A; N/R	Yes; 0.8 m/s
[5]		2.5 m; 1 m^2	5	Adaptive feedback threshold; PD array	OOK	120 kbps	3.43 cm; N/R	Yes; 1 m/s
[9]	Beam-steered OWC	54 m; N/A	1	Optical phased array	Non-return-to-zero (NRZ) and 4-PAM	32 Gbps and 50 Gbps	N/A	No
[10]		3 m; area with diameter of 3 m	1	Spatial light modulator, Infrared camera	QPSK	50 Gbps	N/A; ~1 min	No
[14]		2.5 m; area with diameter of 1.3 m	1	Wavelength routing for beam steering	N/R	10 Gbps	N/A; >115 ms	No
[20]		3 m; 3.14 m^2	1	Camera-based tracking	N/R	10 Gbps	1 mm; 200 ms	No
[22]		2 m; $20 \times 60 \text{ cm}^2$	2	filter-enhanced spatial waveform shaping filters	OOK	2.5 Gbps	4.52-5.41 cm; N/R	Yes; N/R
[23]		1.24 m; $0.5 \times 0.5 \text{ m}^2$	4	Gaussian pulse fitting	N/A	N/A	< 5.3 cm; 30 s	No
[24]		1.15 m; N/R	2	VLP-assisted	N/A	N/A	1.05-6.75 cm; 0.17 ms	No
Proposed design	Beam-scanning OWC	3 m; $50 \times 20 \text{ cm}^2$	1	Tunable laser; Search-and-Communicate Mechanism	OOK	3.4 Gbps	< 3 mm; 16.9 ms	Yes; 255 mm/s

Note: N/A indicates not applicable; N/R indicates not revealed by the author; PAM denotes pulse amplitude modulation; QPSK denotes quadrature phase shift keying; OOK denotes on-off keying.

versa. This challenge is common to most LED-based VLC systems. Yang et al. proposes a combinational light emission approach to enhance data rates, supporting mobility up to 0.8 m/s [4]. However, the reliance on numerous LEDs makes it impractical for precise localization. Alternatively, Jin et al. addresses mobility by introducing a photodetector (PD) array at the receiver end and using an adaptive feedback threshold to demodulate varying signal intensities caused by distance fluctuations between transmitter and receiver [5]. Despite these efforts, the data rate remains limited to 120 kbps, highlighting the difficulties VLC systems face in balancing positioning and communication performance. The comparisons between indoor OWC technologies are summarized in Table I.

In BS-ILC systems, the method of beam steering is a crucial aspect with various methods available for the transmitter. Optical phased arrays (OPAs) can steer beams in one or two dimensions, making them suitable for use in various optical communication and sensing systems [6], [7], [8], [9]. For instance, Lu et al. explored an OPA design utilizing silicon nitride delay lines for two-dimensional (2D) scanning [6], while Raptakis et al.

focused on 3D polymer photonics integrated circuits for beam steering [7]. In sensing applications, additional considerations, such as the reception of reflected light and potential signal loss, are crucial. The use of OPAs in optical wireless communication was demonstrated in [8] and [9], where [8] achieved a communication range of just 3 cm, while [9] proposed a 54 m communication distance and demonstrated channel switching between two receivers spaced 3.5 cm apart horizontally. However, neither study addressed continuous beam alignment on moving objects. Furthermore, [7] and [9] lacked thermal stability analysis, and there is limited information on the manufacturability and scalability of these OPA-based applications for mass production.

Both [10], [11], and [12] utilized liquid-crystal spatial light modulators (LC-SLMs) at the receiver end to improve beam alignment with the PD for communication. However, this approach introduces additional computational complexity at the receiver, suffers from limited response time, making it unsuitable for dynamic scenarios, and results in signal losses due to scattering and diffraction inherent to LC-SLMs. In [13], LC-SLM was successfully used to achieve non-line-of-sight (NLOS) BS-ILC,

but the process of wavefront optimization for beam focusing and steering required approximately 400 seconds to complete.

The system described in [14] and [15] used an arrayed waveguide grating router (AWGR) with 144 or 36 output ports arranged in a 2D fiber array. Each incoming wavelength is directed to a specific fiber in this array, which determines the direction of the emitted beam after it passes through a lens. However, the layout of the output ports is fixed during manufacturing, meaning it cannot be changed later. As a result, fast wavelength switching is necessary for beam steering, with each channel assigned to a specific position. Since there is no overlap between adjacent channels, this setup is not ideal for situations that require dynamic adjustment.

The benefits of beam-steered OWC, including BS-ILC, over VLC, such as high data rates, longer transmission distances, and enhanced privacy, can only be fully realized if the transmitter has the receiver location information. The most widely used localization scheme for this involves cameras, as demonstrated in several studies [11], [16], [17], [18], [19], [20], [21]. These approaches either rely on image recognition algorithms to detect the receiver location or use a combination of an infrared camera on the transmitter and infrared beacon lights near the receiver. While image recognition methods have been tested with monotonous backgrounds, the system robustness needs further evaluation, especially since this method fails in low-light conditions [16], [17]. The beacon light method significantly increases power consumption on the receiver end [11], [18], [19], [20], [21]. Additionally, all these camera-based localization schemes function as separate tracking systems that operate independently of the communication system, adding to the overall complexity.

Multi-light path localization approaches have been explored as potential solutions. Wang et al. introduced a filter-enhanced system that enables continuous and simultaneous positioning and data communication by applying spatial waveform-shaping filters to the transmitters [22]. This method allows direct received signal strength (RSS)-based positioning using communication signals without requiring dedicated positioning resources but necessitates the addition of more transmitters. Another approach demonstrated an experimental time-difference-of-arrival (TDoA)-based indoor positioning system, achieving a 95th percentile positioning accuracy of 5.3 cm in a 50×50 cm² grid area at a distance of 1.24 m [23]. However, its high data acquisition latency of 30 seconds makes it unsuitable for mobile scenarios. Additionally, VLP-assisted laser-based OWC has been employed to support end-user mobility at a distance of 1.15 m [24]. This system requires a PD array at the receiver end for localization. While simulations indicate it can track a target receiver moving at speeds up to 10 m/s, the system actual data rate has not been disclosed, leaving its practical communication performance uncertain.

This highlights a key challenge for many existing systems: the positioning and communication functions are often not integrated within a compact design. To overcome the limitations of existing optical wireless communication approaches, we propose an O-ISAC system—a two-dimensional (2D) beam scanning architecture that employs a single light source to achieve dynamic localization and continuous alignment with a moving

target. The system integrates sensing and communication in a compact design. Experimental validation demonstrates its ability to accurately track targets in various motion trajectories, including horizontal, vertical, and diagonal movements, with an average tracking error of less than 3 mm within the receiver's effective area. Furthermore, the system maintains a stable 3-meter optical link while the target is in motion, achieving a data rate of 3.4 Gbps.

As illustrated in Fig. 1, the signal originates from the core network and is transmitted via free-space optics to various indoor environments. This O-ISAC framework is well-suited for mobile indoor IoT applications, such as wireless data delivery to virtual reality (VR) devices or real-time communication with cooperative mobile robots. In the context of data centers, the system architecture supports communication between server racks by utilizing ceiling-mounted silver-coated mirrors. Each rack is equipped with a movable transceiver, which dynamically adjusts its orientation to establish a line-of-sight link via ceiling reflections. This configuration enables communication between racks, even when direct paths are initially obstructed by intervening equipment. The mobility of the transceivers also facilitates more complex routing strategies, enhancing system scalability and throughput. Additionally, the cable-free design reduces spatial constraints, minimizes thermal load, and improves overall cooling efficiency. In collaborative robotics environments, the same O-ISAC system can enable real-time coordination and data sharing between mobile robotic units, further underscoring its versatility in dynamic indoor settings.

II. SYSTEM MECHANISMS: 2D BEAM SCANNING, PASSIVE LOCALIZATION, AND SEARCH-AND-COMMUNICATE

This section begins by explaining the working principles of the 2D beam scanning system. Fast-axis scanning is achieved through the integration of a tunable micro-electro-mechanical system (MEMS)-vertical-cavity surface-emitting laser (VCSEL) and a transmission grating, while the slow axis is controlled by a one-dimensional (1D) galvanometer (galvo) mirror. Following this, we introduce the passive localization scheme employed in the system. This scheme leverages the characteristics of a corner cube structure, enabling precise receiver tracking without requiring active participation from the receiver. Next, we present the coaxial optical sensing system, where the transmission and detection of light are integrated at the transmitting end. This system facilitates the localization of the receiver, allowing for position determination using reflected light. This section further details the Search-and-Communicate mechanism, which operates in two phases. The process commences with a Search phase, during which the system acquires and processes data to accurately track the target position. Once localization is achieved, the system transitions into the Communication phase, where the optical link is established and maintained.

A. 2D Beam Scanning Mechanism

The integration of tunable MEMS-VCSEL and the transmission grating has become a common technique in swept-source light detection and ranging (LiDAR) systems, as referenced in

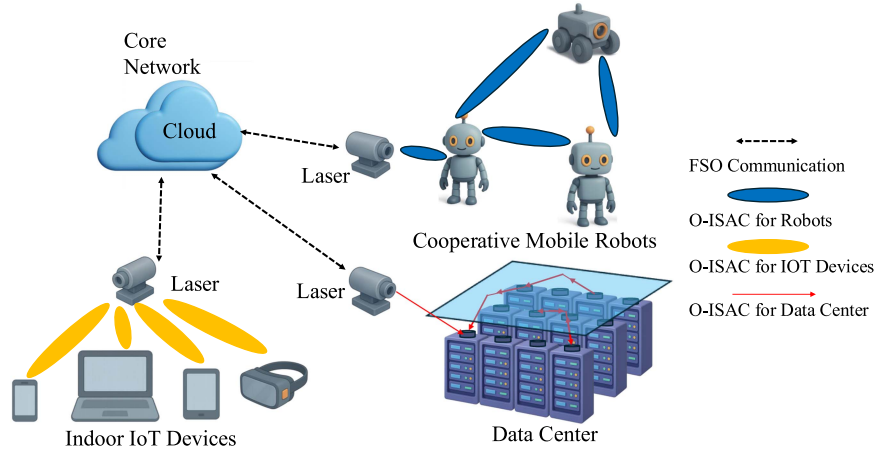


Fig. 1. IoT scenarios for the proposed Optical Integrated Sensing and Communication (O-ISAC) system.

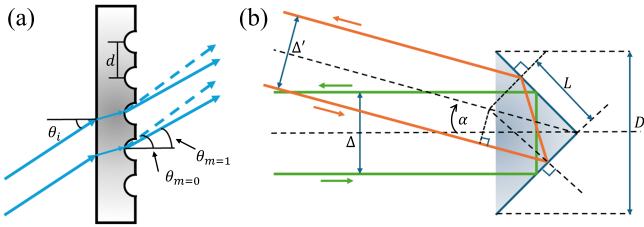


Fig. 2. (a) The principle of transmission grating. (b) 2D view of ray paths in optical corner cube reflector.

[25], [26], [27]. The process begins with the tunable MEMS-VCSEL, applying a voltage to an electro-optic material shifts the refractive index or changes the spacing of the cavity mirrors. This leads to a shift in the lasing wavelength or referred to as frequency swept. The relation between the applied voltage V and the output wavelength λ is generally linear over a limited range as

$$\lambda(V) = \lambda_0 + \alpha V, \quad (1)$$

where λ_0 represents the center wavelength when the voltage $V = 0$, and α is the tuning coefficient indicating how sensitive the wavelength is to changes in voltage. The transmission grating disperses light based on its wavelength, meaning different wavelengths are diffracted at different angles as shown in Fig. 2(a). The diffraction angle θ for a given λ is described by the grating equation

$$m\lambda = d(\sin \theta_m - \sin \theta_i), \quad (2)$$

where m is the diffraction order (typically $m = 1$ first-order diffraction), d represents the grating spacing, defined as the distance between adjacent grooves. Physically, the m corresponds discrete directions into which light is diffracted by the grating. The first order ($m = 1$) represents the primary diffracted beam, which carries the most significant portion of the diffracted light energy and exhibits the desired wavelength-dependent angular dispersion used for beam steering in our system. Higher diffraction orders ($m = 2, 3$, etc.) would represent additional,

more widely dispersed beams, while the zeroth order ($m = 0$) corresponds to undiffracted light that passes straight through the grating without angular dispersion. And θ_i is the incident angle of the incoming laser beam. As the λ of the VCSEL changes due to tuning voltage adjustments, θ also changes. For a fixed grating and incident angle, this causes beam steering across different angles. The angular shift $\Delta\theta$ for a $\Delta\lambda$ can be derived from the (2) as

$$\Delta\theta = \frac{\Delta\lambda}{d \cos \theta}. \quad (3)$$

By combining the (1) and (3), the voltage-to-angle relationship can be expressed as (4). This shows that adjusting the voltage on the tunable VCSEL controls the steering angle θ of the outgoing light after diffraction

$$\Delta\theta = \frac{\alpha \Delta V}{d \cos \theta}. \quad (4)$$

For the slow-axis scanning uses the 1D galvo mirror, which moves the beam horizontally by adjusting its angle based on the applied control signal. When combining the fast and slow axis mechanisms, the result is a 2D scanning pattern, and the 2D beam scanning procedure is as follow

- 1) In the vertical fast-axis scanning, the tunable laser emits light at different wavelengths, and the transmission grating disperses the light, directing each wavelength at a specific angle. As the wavelength changes, the outgoing light beam shifts vertically, sweeping from top to bottom across the target. This rapid vertical scan forms one complete vertical line at a time.
- 2) Simultaneously, the galvo mirror controls the slow-axis motion, which moves the beam horizontally. After each vertical scan is complete, the galvo mirror adjusts the beam to the right (or left), shifting the scan to the next horizontal position. This creates a stepping motion from left to right. The system continues this process of vertical scanning followed by horizontal stepping, forming a snake-scan across the predefined area.

B. Passive Localization Scheme

To minimize system complexity and significantly lower power consumption of receiver end, Koonen et al. explored a passive localization scheme utilizing a retroreflector based on the optical corner cube [14]. The working principle of the corner cube retroreflector (CCR) relies on the phenomenon where light rays entering the CCR are reflected long the same path from which they arrived. This behavior results from the geometric structure of the CCR, which consists of orthogonal reflecting surfaces. As shown in Fig. 2(b), the displacement Δ is influenced by the entry point of the incoming ray into the CCR and the angle of incidence α with respect to the CCR's main axis

$$\Delta = 2L \cdot \sin\left(\frac{\pi}{4} - \alpha\right), \quad (5)$$

the maximum displacement $\Delta_{\max}$ is proportional to the entry aperture D of the CCR and can be described by

$$\Delta_{\max} = D \cdot \sqrt{2 \cdot \sin\left(\frac{\pi}{4} - \alpha\right)}. \quad (6)$$

We present a coaxial optical sensing system designed for dynamic localization of a retroreflector within a 2D plane. The system integrates both the transmission and detection of light at the transmitter side, using a combination of optical components and balanced photodetectors (BPDs) for tracking. The process begins with the emission of light from a collimator. Before this light is transmitted, BPD 1 is employed to measure the power of the outgoing beam, providing a reference value for later comparison. The reference power $P_{ref}(t)$ serves as a baseline against which the reflected light is compared. Once the transmitted light reaches the retroreflector, the reflected light is collected again by the same collimator. The reflected light is then passed through an optical circulator to ensure proper routing and is measured by BPD 2. The power of the reflected signal $P_{meas}(t)$ is recorded, allowing for a comparison with the $P_{ref}(t)$. The system operates on the principle that the difference between the reference and reflected signals $\Delta P(t)$ will reveal whether the retroreflector is present within the scanning area. If the reflected light is successfully detected, $\Delta P(t)$ will exceed a predefined threshold, indicating the presence of the retroreflector. As introduced earlier, the 2D beam-scanning mechanism dynamically steers the light beam across a predefined area. As the light scans the area, the reference and measured signals are recorded over time by an oscilloscope.

Since the beam scanning occurs as a temporal sequence, the light progressively scans the entire scene. The oscilloscope captures the time-series data for both the reference and reflected signals. This data is used to reconstruct the retroreflector's position within the 2D plane afterward. To ensure accurate detection, the system parameters, such as lens size and scan speed are designed to accommodate the $\Delta_{\max}$ between the transmitted and reflected light paths. This guarantees consistent detection of the retroreflector across the scanning area.

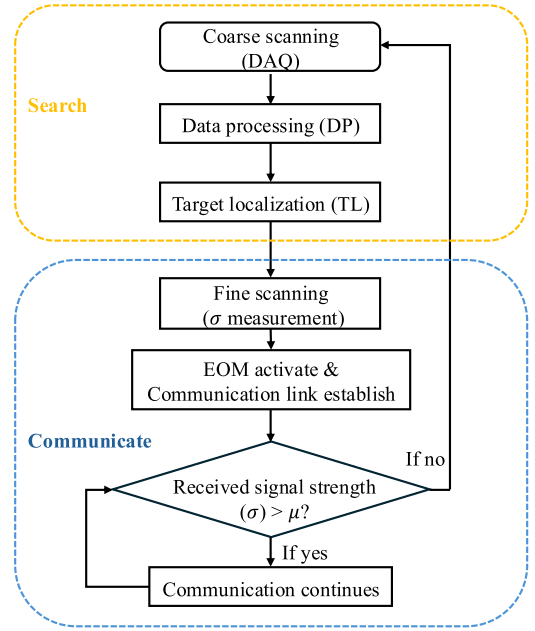


Fig. 3. Flowchart of Search-and-Communicate mechanism.

C. Search-and-Communicate Mechanism

The mechanism operates in two phases: Search phase and Communication phase as given in Fig. 3. In the Search phase, it begins with coarse scanning of a predefined area. The oscilloscope captures the measured and reference values over a duration of data acquisition (DAQ) for the frames. Then the data is transferred to a processing unit for data processing (DP). During this step, the consecutive data values, initially in the time domain, are organized into a matrix format to enable vectorized calculations. This structured data is then analyzed using a target localization (TL) function to determine the trajectory of the moving object, forming the Search phase.

The TL function processes consecutive frames, denoted as F_t where $t = 1, 2, \dots, N$ represents the frame number. The algorithm uses a localization window defined by dimensions $W = [h_w, w_w]$, where h_w and w_w are the height and width of the window, respectively. The input parameters include τ , which is the minimum pixel intensity difference required for object detection, and the search factor α , the percentage of the frame that will initially be searched. The output of the algorithm is a matrix T containing the target coordinates $[x_t, y_t]$ for each frame t . Initially, each frame F_t is read, with dimensions $H_f \times W_f$. An initial search area is defined with dimensions $H_s = \alpha H_f$ and $W_s = \alpha W_f$, centered around $(x_c, y_c) = (\frac{W_f}{2}, \frac{H_f}{2})$, where x_c is the horizontal coordinate of the center of the frame and y_c is the vertical coordinate of the center of the frame. For each frame F_t , the grayscale version of the frame, G_t is computed, and the average pixel intensity of the entire frame is denoted as $\bar{I}_b$, which serves as the reference background intensity. This is given by the formula

$$\bar{I}_b = \frac{1}{H_f W_f} \sum_{i=1}^{H_f} \sum_{j=1}^{W_f} G_t(i, j). \quad (7)$$

Next, a scanning window W is moved across the search area starting from $(x_c - \frac{W_s}{2}, y_c - \frac{H_s}{2})$ to $(x_c + \frac{W_s}{2}, y_c + \frac{H_s}{2})$. For each window position (x_w, y_w) , the average intensity $\bar{I}_w$ of the pixels within the window W is calculated

$$\bar{I}_w = \frac{1}{h_w w_w} \sum_{i=x_w}^{x_w+w_w-1} \sum_{j=y_w}^{y_w+h_w-1} G_t(i, j), \quad (8)$$

and the absolute difference ΔI between the $\bar{I}_w$ and $\bar{I}_b$ is computed: $\Delta I = |\bar{I}_w - \bar{I}_b|$. If $\Delta I > \tau$, the position (x_w, y_w) of the window is considered a candidate for the target. The corresponding $\Delta I_{\max}$ is selected as the target's location (x_t, y_t) for the current frame F_t . For $t \geq 2$, the α is decreased to a predefined smaller value α_{narrow} . The narrow search area is defined as a region centered around (x_{t-1}, y_{t-1}) from the previous frame $H_s = \alpha_{\text{narrow}} H_f$ and $W_s = \alpha_{\text{narrow}} W_f$, the search area spans $(x_{t-1} - \frac{W_s}{2}, y_{t-1} - \frac{H_s}{2})$ to $(x_{t-1} + \frac{W_s}{2}, y_{t-1} + \frac{H_s}{2})$. The coordinates (x_t, y_t) are smoothed using the previous coordinates, with the smoothing factor γ

$$(x_t, y_t) = \gamma (x_t, y_t) + (1 - \gamma) (x_{t-1}, y_{t-1}), \quad (9)$$

and these coordinates are added to matrix

$$T = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_N & y_N \end{bmatrix}. \quad (10)$$

If $\Delta I_{\max} \leq \tau$, the algorithm increases α by 2 and repeats the process until localization is completed for all frames. After target localization is completed, the 2D beam scanning system switches from coarse to fine scanning mode, activating the Communication phase. This transition is achieved by adjusting the voltage of the tunable laser source and modifying the control signal for the galvo mirror. Once the system enters the Communication phase, the electro-optic modulator (EOM) initiates signal transmission, establishing the communication link. During Communication phase, two BPDs continuously monitor the accumulated reflected signal strength σ , ensuring it remains above a predetermined threshold μ . This μ is defined as the minimum signal strength required to ensure that the projected communication window fully covers the effective area of the photodetector at receiver. If σ falls below μ , indicating potential misalignment or a loss of coverage, the system reverts to the search phase, performing the coarse scanning. The process of target localization is repeated to reacquire the receiver position, after which communication resumes.

III. EXPERIMENTAL SETUP, RESULT AND DISCUSSION

The experimental setup is shown in Fig. 4(a). The frequency-swept light is generated by the 1550-nm tunable MEMS-VCSEL (BW10-1550-T-PSFA, Bandwidth10), utilizing a high contrast grating (HCG). It attains a peak optical output power of 3 dBm and a sidemode suppression ratio of 40 dB [15]. In this experiment, the tunable MEMS-VCSEL outputs an optical power of 2 dBm. An arbitrary function generator (AFG, FG3011C, Tektronix) supplies the modulation voltage required for the

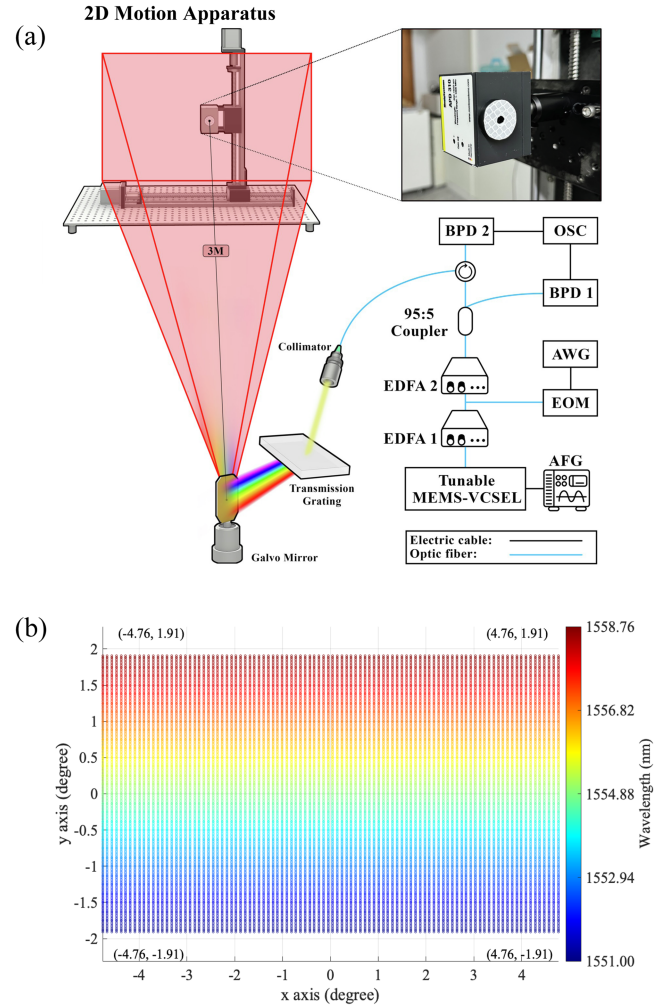


Fig. 4. (a) Schematic diagram of the dynamic localization and communication system with 2D beam scanning. EDFA: Erbium-doped fiber amplifier; AFG: arbitrary function generator; AWG: arbitrary waveform generator; EOM: electro-optic modulator; OSC: oscilloscope; BPD: balanced photodetector. (b) The coarse scanning pattern of the 2D beam scanning system, uses color variations to show changes in wavelength. The x and y axes represent the angular displacement from the transmitter, covering a predefined $20 \times 50 \text{ cm}^2$ area located 3 meters away.

laser frequency sweep. This voltage ranges from 1 V to 18 V at a modulation frequency of 20 kHz, which corresponds to a wavelength sweep from 1558.76 nm to 1551.00 nm, providing a tuning range of 7.76 nm. The linear frequency-swept laser from the MEMS-VCSEL is amplified using an erbium-doped fiber amplifier (EDFA 1, EDFA-D-O-23, fiber-photonics) with a gain of 11.2 dB. The amplified light is modulated by an EOM (MZM, FTM7938EZ, Fujitsu), with the modulated signals generated by an arbitrary waveform generator (AWG, AWG7101, Tektronix). After further amplification by EDFA 2 with a gain of 11 dB, the light is directed into a 95/5 optical coupler. At this stage, 95% of the laser light is transmitted through a circulator and collimator (F280APC-1550, Thorlabs), achieving an optical power of 22.78 dBm before projected onto the target as part of the 2D beam scanning system. The remaining 5% light is collected by BPD 1 (PDB570C, Thorlabs), serving as the reference value for subsequent data processing. When the reflected light from the

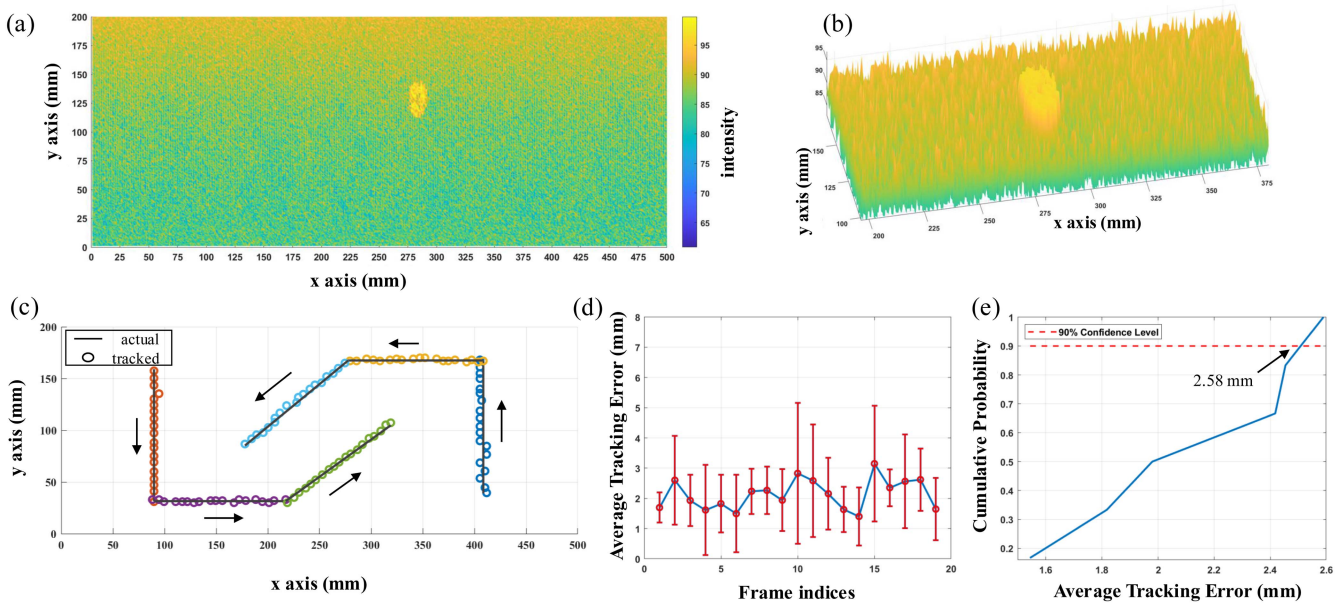


Fig. 5. (a) Front view displaying intensity changes with an accompanying color bar; (b) Zoomed-in view of the target receiver; (c) Illustration of six distinct trajectories, where the circle marks the center of the tracked window, and the straight lines represent the actual trajectories; (d) Graph showing the average tracking error and standard deviation across all trajectories; (e) Cumulative Probability of tracking errors.

target is received by the 2D beam scanning system, it is directed back through the circulator and detected by BPD 2 (PDB570C, Thorlabs) as the measured value.

Once the frequency swept laser exits the collimator and enters free space, it is combined with the transmission grating, acting as the fast-axis scanning, disperses the light vertically based on its wavelength at a modulation rate of 20 kHz. The resulting maximum diffraction angle is ± 1.91 degrees at 3 meters. For the slow scanning axis, the galvo mirror operates at a scanning frequency of 40 Hz, producing a horizontal scanning range of 9.46 degrees at the same distance. By design, the 2D beam-scanning mechanism ensures Class 1 eye safety by distributing optical power over a large area and minimizing dwell time per point. While the static beam poses a hazard (354 mW/cm^2), active scanning reduces effective irradiance to 2.55 mW/cm^2 —far below the 100 mW/cm^2 MPE. This architecture, combined with measured receiver-plane power ($121 \text{ }\mu\text{W}$), guarantees compliance with ANSI Z136.1, *American National Standard for Safe Use of Lasers*, under normal operation. As a result, this 2D beam scanning mechanism covers the target movement across a rectangular area of 20 cm in height and 50 cm in width. The scanned pattern, as shown in Fig. 4(b), illustrates the 2D beam scanning process, with the color bar representing the frequency sweep from 1551.00 nm to 1558.76 nm. The four extremum points are also marked by their corresponding deviation angles from the center, as (± 4.76 degrees, ± 1.91 degrees). Although the system performs a total of 500 vertical scans, only 100 are displayed for clarity.

To evaluate the performance of the dynamic target localization system, a 2D target motion apparatus was constructed at 3 m from the transmitter. This apparatus consists of two mechanical sliding rails (FSK40, FUYU) with a maximum movement speed of 255 mm/s. One rail functions as the horizontal axis, while the

second rail is mounted on top of the sliding component of the first, enabling vertical motion. By simultaneously controlling both sliding rails, more complex motion patterns, including diagonal movement, are achieved. An avalanche photodetector (APD, APD310, MenloSystems), with a frequency range of 1-1800 MHz, is mounted on the sliding component of the vertical rail. A ring of retro-reflective material is affixed around the APD's effective area as illustrated in the inset of Fig. 4(a). The retro-reflective ring has an outer diameter of 25 mm and an inner diameter of 5 mm.

Assessing the system localization capability involved analyzing the reconstructed scene at the receiver end, as shown in Fig. 5(a). The reconstruction data was collected while the target moved downward vertically at a speed of 255 mm/s. By deploying 200 horizontal scans and 500 vertical scans over the predefined $20 \times 50 \text{ cm}^2$ area, as described in the experimental setup, the system resolution is approximately 200×500 pixels. The reconstruction displays the intensity distribution across the scanned area, with a color bar indicating signal strength variations. A more detailed view of the target receiver is provided in Fig. 5(b). The average calculated area of the reconstructed retro-reflector ring is 377 mm^2 , with a height of 32 mm and a width of 16 mm, which is smaller than the designed retro-reflector ring size of 490 mm^2 (with the center filled). This discrepancy is due to inconsistent scanning speeds across the two axes. Despite this inconsistency, the retro-reflector ring retains a stable reconstructed shape at various positions within the predefined area.

In addition to static reconstruction, we implemented the Search-and-Communicate mechanism to dynamically follow the moving target, as detailed in the methodologies. The time taken for data acquisition for each frame is 12.5 ms. The processing time for TL for the first frame is 1.3 ms, as the initial

TABLE II
LATENCY EVALUATION FOR KEY PROCEDURES

Description	Duration (ms)
Time taken for data acquisition (DAQ) per frame	12.5
Processing time of target localization (TL) w/ search factor α , (for the 1st frame)	1.3
Average processing time of TL w/ search factor α_{narrow} (for the 2nd and subsequent frames)	0.6
σ measurement	2.5

search factor α is set to 0.5. This guarantees a larger scanning window to localize the target in the first frame. For the second and subsequent frames, the average TL processing time reduces to 0.6 ms, as α_{narrow} is updated to 0.1, shrinking the scanning window. These latency details for each step are summarized in Table II. Fig. 5(c) illustrates the trajectories of six different motions within the scene. Each line represents the target moving in a specific direction for 0.5 seconds at a speed of 255 mm/s. Since the DAQ duration per frame is 12.5 ms, and an additional 12.5 ms is allocated for DP and TL, each complete cycle of DAQ, DP, TL, and idle time requires 25 ms. Over the 0.5-second interval, the system processes 20 frames, with 19 frames used in this case. The latency evaluations for each step were conducted on the MATLAB R2023b using a system equipped with an AMD Ryzen 9 5900HX CPU. The arrows in Fig. 5(c) indicate the movement directions for each trajectory, covering horizontal, vertical, and diagonal motion. The circular symbol marks the center of the tracked position, as the TL function uses the scanning window to identify the tracked position. As shown, the tracked trajectories fit the actual trajectories (represented by straight lines) consistently. Fig. 5(d) illustrates the average tracking error between the tracked and actual trajectories during the search phase. The results show that the overall average tracking error remains below 3 mm, which is approximately 8 times smaller than the outer diameter of the retro-reflector ring. The error bars at each frame index represent the standard deviation of the tracking error, indicating the consistency of the tracking performance. Additionally, Fig. 5(e) demonstrates the cumulative tracking error distribution. The results indicate that 90% of the cumulative tracking errors of the TL function are within 2.58 mm, further showcasing the precision of the tracking mechanism.

After the TL is complete, the system transitions to Communication phase, beginning with the fine scanning stage. In this stage, the scanning window is reduced by adjusting the voltage applied to the tunable MEMS-VCSEL and modifying the control signal sent to the galvo mirror. The horizontal scanning frequency is increased to 200 Hz, and the scanning window size is reduced to $35 \times 35 \text{ mm}^2$. The fine scanning window dynamically follows the coordinates provided by the TL function. To ensure the communication link remains stable during movement, the σ measurement is taken every 2.5 ms.

The communication process begins with the generation of a 5000-bit pseudo-random binary sequence (PRBS), including a preamble of 500 zeros followed by 4500 random bits. This sequence undergoes on-off keying (OOK) modulation, where each bit in the sequence is mapped to either a high or low light intensity (1 or -1). This approach aligns with the recommendations outlined in the *IEEE Standard for Local and Metropolitan Area Networks—Part 15.7: Short-Range Optical Wireless Communications* (IEEE Std 802.15.7-2018). The OOK-modulated signal is then pulse-shaped, with a raised cosine filter, to limit bandwidth while maintaining signal integrity. The pulse-shaped signal is emitted by the AWG at different frequencies and modulated onto the tunable MEMS-VCSEL light source via the EOM. The transmitted signal $s(t)$ can be expressed as

$$s(t) = \sum_{n=0}^{N-1} a_n p(t - nT), \quad (11)$$

where $a_n \in \{0, 1\}$ represents the binary data (with $a_0, a_1, \dots, a_{499} = 0$ as the preamble), $p(t)$ is the pulse-shaping filter, and T is the symbol duration.

At the receiver end, the signal is captured by the oscilloscope, and the following steps are applied

- 1) The 500-bit preamble of zeros is employed to synchronize the transmitted and received signals. The cross-correlation function is given by

$$C(\tau) = \int r(t) s(t + \tau) dt, \quad (12)$$

where $r(t)$ is the received signal and $s(t)$ is the transmitted signal. The time shift τ that maximizes $C(\tau)$ corresponds to the correct alignment of the $r(t)$ with the $s(t)$, facilitating synchronization.

- 2) $r(t)$ is match-filtered to maximize the signal-to-noise ratio (SNR), correlating the received waveform with the transmitted pulse shape. The output of the match filter is

$$r_{\text{filtered}}(t) = \int r(\tau) p(t - \tau) d\tau. \quad (13)$$

- 3) To mitigate inter-symbol interference (ISI), the received signal undergoes feed-forward equalization (FFE), using an adaptive filter with coefficient c_n . This compensates for the channel distortion, producing

$$y(t) = \sum_{n=0}^{N-1} c_n r_{\text{filtered}}(t - nT). \quad (14)$$

- 4) After equalization, the signal is resampled to ensure it matches the original data rate of the transmitted signal. The resampled signal $r_{\text{resampled}}(nT)$ is compared to a threshold V_{th} to demodulate the OOK signal

$$\hat{a}_n = \begin{cases} 1 & \text{if } r_{\text{resampled}}(nT) > V_{th} \\ 0 & \text{if } r_{\text{resampled}}(nT) \leq V_{th} \end{cases}. \quad (15)$$

- 5) The previous steps ensures that the bitstream is correctly aligned at the receiver before bit error rate (BER) calculation is applied to the remaining 4500 bits. The BER is calculated by comparing the demodulated sequence $\hat{a}_n$

with the transmitted sequence a_n , as follow

$$BER = \frac{1}{N} \sum_{n=1}^N |a_n - \hat{a}_n|, \quad (16)$$

where N represents the total number of bits transmitted, excluding the preamble.

Due to the sensitivity distribution of the APD, with the highest sensitivity at the center, the scanning light produces a bumpy signal at the receiver, as shown in Fig. 6(a). The preamble zeros highlight the gaps between data sequences. Approximately 8 data sequences are captured per vertical scan, but only the three highest energy sequences are used for further processing due to lower SNR in the others. After the FFE equalization, the scattering diagram in Fig. 6(b) displays a total of 5000 data points. Roughly 2500 points are clustered in the upper half, and 2500 in the lower half, corresponding to OOK '1' and '-1' symbols. A clear gap separates these two clusters, indicating a distinct threshold between high and low light intensities. This separation reflects effective mitigation of ISI and noise. In the performance evaluation of the communication system, the BER was plotted against data rates ranging from 3.1 to 3.5 Gbps, as shown in Fig. 6(c). The results indicate a gradual increase in BER as the data rate approaches 3.5 Gbps, where it reaches the hard-decision forward error correction (HD-FEC) limit. Furthermore, to assess the system communication feature in a dynamic environment at a data rate of 3.4 Gbps, we tested its performance on a moving target at different positions within a $50 \times 20 \text{ cm}^2$ rectangular area. The target moved at a speed of 255 mm/s, and we selected three vertical levels within the area: 9 cm, 12 cm, and 19 cm, each denoted by different colors in Fig. 6(d). At each of these levels, we measured the BER at five positions, spaced 10 cm apart. In total, 15 positions were tested, and the BER for all positions remained below the HD-FEC limit of 3.8×10^{-3} , demonstrating the system capability to maintain reliable communication even with a moving target.

As highlighted in the introduction, existing VLC systems have struggled to achieve data rates comparable to the 3.4 Gbps demonstrated in our experiment, with positioning errors typically remaining in the centimeter range. These limitations arise from the inherent bandwidth constraints of LEDs, which cap data rates and restrict localization precision. Alternative beam-steering methods, such as OPAs, LC-SLMs, or 2D fiber array, face additional challenges, particularly in their inability to maintain continuous communication links while the target receiver is in motion.

In our system, the 3.4 Gbps data rate is achieved using a single light source and a simple OOK modulation format. This highlights the simplicity and efficiency of the setup. Additionally, deploying advanced modulation techniques, such as quadrature amplitude modulation (QAM), could further increase data rates, leveraging higher spectral efficiency.

Another critical challenge in laser-based OWC systems is the receiver APD. While high-bandwidth APDs enable faster data acquisition, their small effective detection areas—such as the 0.04 mm diameter used in this work—make precise beam

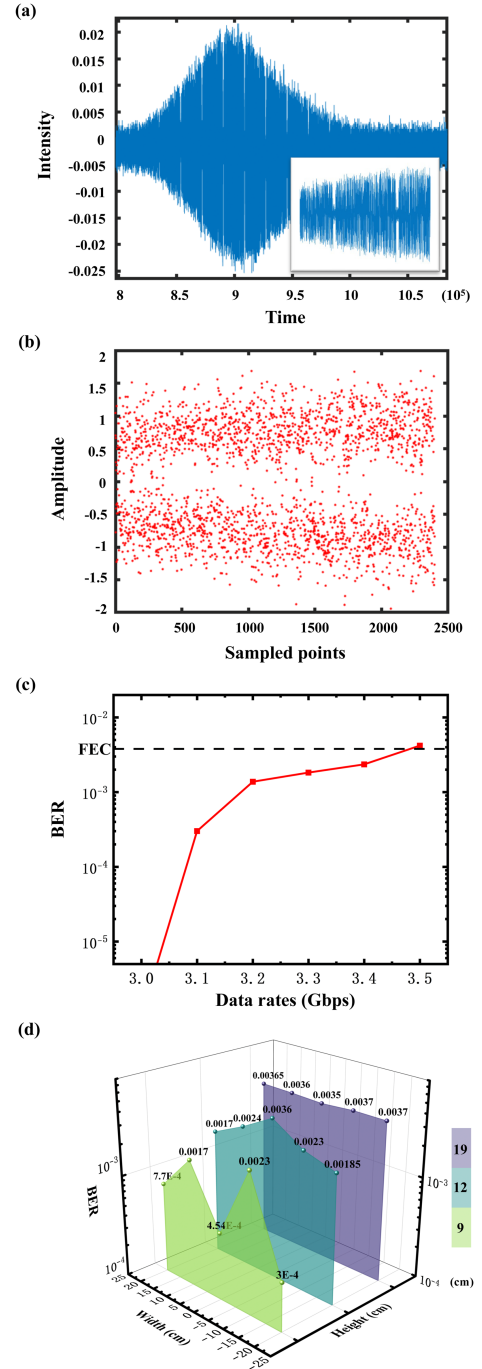


Fig. 6. (a) The received signal at the APD. (b) The scatter plot shows separation between '1' and '-1' symbols after FFE. (c) The result of BER versus data rate for a static target. (d) BER measurements at 15 positions with a moving target at a data rate 3.4 Gbps.

alignment challenging, especially over the 3 m transmission distance.

IV. CONCLUSION

In this paper, we presented the 2D beam scanning system using a single light source for dynamic target localization and continuous beam alignment for simultaneous communication. This system exemplifies the concept of O-ISAC, wherein a

shared optical path supports both functions in a compact, efficient architecture. The 2D beam steering mechanism utilizes a diffraction grating and tunable MEMS-VCSEL for the fast axis, with a galvo mirror controlling the slow axis. By integrating a passive localization scheme based on the corner cube structure, along with the Search-and-Communicate mechanism and custom digital signal processing techniques at both the transmitter and receiver, the system provides a reliable solution for maintaining a free-space communication link. Experimental evaluations confirmed the system localization capability, achieving a resolution of 200×500 pixels over a 20×50 cm² area with an average reconstructed retro-reflector area of 377 mm². The TL function exhibited an average tracking error below 3 mm. Communication performance tests demonstrated a data rate of 3.4 Gbps with bit error rates (BER) below the HD-FEC limit of 3.8×10^{-3} , even while the receiver is in motion over a 50×20 cm² area. These results validate the system potential across diverse indoor IoT scenarios, including dynamic wireless communication among data center racks, cooperative robotics, and immersive applications such as VR. By enabling cable-free, high-speed, and precisely aligned optical links, the proposed O-ISAC system addresses key challenges in bandwidth demand, spatial flexibility, and energy efficiency in modern indoor environments.

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